FET DC bias

Cheat sheet · Day 17 · Mon 9 Nov · pack E10

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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FET DC BIAS  (I_G = 0, I_D = I_S, no drop across R_G; work in mA and kOhm)
JFET/D-MOS:  I_D = IDSS(1 - VGS/VP)^2      E-MOS: I_D = k(VGS - VT)^2,
             VGS = VP(1 - sqrt(ID/IDSS))          k = ID(on)/(VGS(on) - VT)^2
Sketch: VGS = 0, 0.3VP, 0.5VP, VP  ->  ID = IDSS, IDSS/2, IDSS/4, 0
Bias lines:  fixed   VGS = -VGG
             self    VGS = -ID*RS
             divider VGS = VG - ID*RS,  VG = R2*VDD/(R1+R2)
             CG/-VSS VGS = VSS - ID*RS   (VS = -VSS + ID*RS)
             E-MOS drain feedback  VGS = VDS = VDD - ID*RD
Output:  VD = VDD - ID*RD,  VDS = VD - VS
Method:  bias line -> sub into device eqn -> quadratic -> keep root with
         VP <= VGS <= 0 (JFET), VGS >= VP (D-MOS, can be +), VGS > VT (E-MOS)
Design:  target VD -> ID = (VDD-VD)/RD;  ID -> VGS (invert);
         VS = VG - VGS;  RS = VS/ID;  RD = (VDD - VD)/ID
ALWAYS CHECK SATURATION: VDS >= VGS - VP  (E-MOS: VDS >= VGS - VT)
  Fails (or VDS < 0) -> ohmic region, Shockley Q-point not valid; say so.
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